Professional Certificate of Competency in Electrical Power System Fundamentals for Non-Electrical Engineers

Webinar 5

Module 6: Faults, Fuses and Circuit Breakers

Module 6 Topics

- Electrical faults
- Fuses
- Isolation, switching and tripping concepts
- Medium voltage circuit breakers

Fault Types

Fault is an abnormal occurrence on a power system.

Active faults can be classified into *solid* and *incipient faults*.

- **Solid faults (e.g. short circuit):** Immediate, complete breakdown of insulation causing:
  - High fault currents/energy
  - Danger to personnel
  - High stressing of all network equipment due to heating and electromechanical forces and possibility of combustion
  - Dips on the network voltage affecting other parties
  - Faults spreading to other phases

Fault Types

- **Incipient faults (e.g. partial discharge):** A fault that takes a long time to develop into a breakdown of insulation caused by:
  - Partial discharge currents
  - These faults normally become solid faults in time

Fault Types

- **Passive faults:** These are conditions that will cause faults in the long term due to cumulative effects, such as:
  - Overloading (over heating insulation)
  - Overvoltage (over stressing insulation)
  - Under frequency
  - Power swings (damages generators)

Consequences of Faults

- Heavy currents damage equipment extensively:
  - Danger of fire hazard
- May lead to long outage times to restore back:
  - Lost production and lost revenue
- Heavy currents through earth bonded enclosures give rise to high touch potentials:
  - Dangerous to human life
- Large fault currents are more hazardous in atmospheres having ignitable gases:
  - Explosion hazard

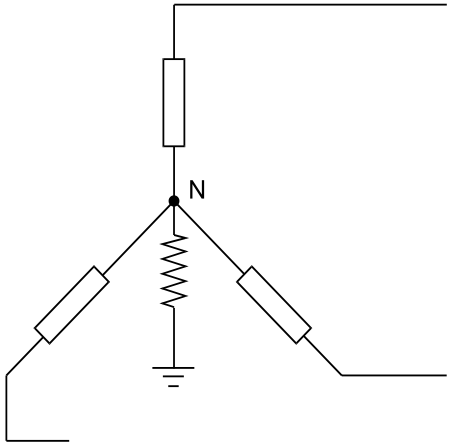
Solutions

Phase segregation (separating phases far apart):

- Mitigate against phase-to-phase faults.

Resistance earthing:

- To lower earth fault currents
- Value can be chosen during design stage to limit current to desired value



Benefits

- Fault damage now minimal
- Lower outage times
- Touch potentials kept within safe limits
- Lower magnetic or thermal stresses imposed on plant during fault
- Transient overvoltage limited

Equipment Selection and Isolation

- Equipment selection so that they can withstand the flow of fault current for a specific duration (1 sec or 3 sec)
- Ensure isolating the fault well within this duration

Limiting Earth Fault Currents

- Earth faults currents depend on the resistance in the neutral circuit.
- Add resistance in the source neutral so that the fault current is limited.

Main Protection Devices

- It is necessary to have equipment that are capable of sending the faults and open the circuit when preset fault currents are exceeded
- **Protection devices** capable of automatic isolation under faults:
  - Fuses
  - Circuit breakers used with relays and tripping devices

Fuse: The Basic Protection Device

A **fuse** is the most basic of all protective devices and performs all the protection functions normally obtained by several devices, i.e. detection, isolation/opening, etc.

- A fuse is a conducting material with maximum current rating and melts when the current exceeds this value thereby opening the circuit under overload and fault currents which otherwise could damage the equipment connected.
- Common fuse material is copper or tinned copper or silver with varying thicknesses specified in wire gauges for different current ratings.

Fuses - Types

Rewirable type:

- Fusible wire: It is open to abuse, incorrect rating used to keep circuit in.

Cartridge type:

- Silver element enclosed in a barrel of insulating material (sometimes filled with quartz sand). Most common use.

Cartridge Type Fuses

Advantages:

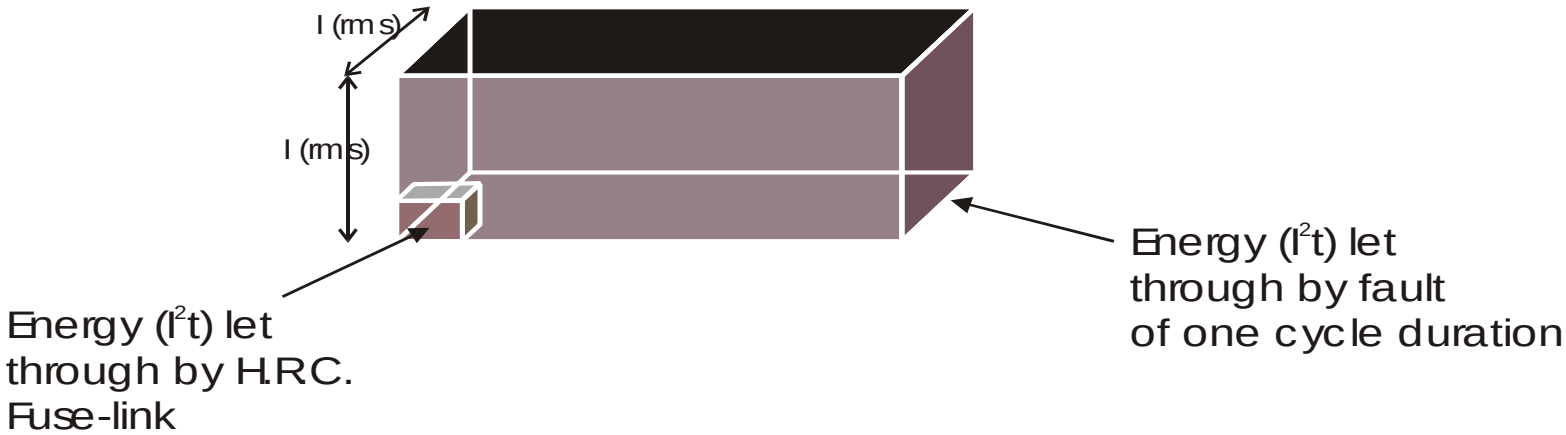
- Fault energy contained within an insulating tube
- Sealed hence does not deteriorate as fast as open type
- Better grading possible
- Quartz sand absorbs energy and melts across ionised metal path to quench arc
- Faster and can handle very high currents up to 100 kA
- Also referred as High Rupturing Capacity (HRC) fuse as they can be designed for protecting a circuit carrying hundreds of amperes

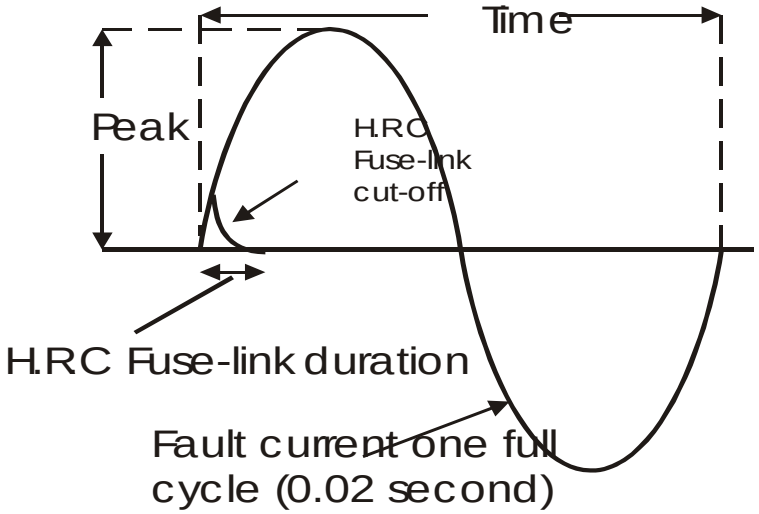
Energy “Let Through”

Energy **let through** basically refers to the energy let into the circuit till fusing and its value is proportional to I^2 x t.

- Fuses can limit this let through energy from the source by fusing very quickly - usually under one fourth of an ac cycle = ¼ x 20 m s = 5 m sec
- In comparison, circuit breakers usually take longer time (up to 10 cycles) → 10 x 20 ms = 200 m sec to open
  - I.e. 40 times more energy is released into the fault compared to a fuse which breaks the current flow in ¼ cycle

Energy "Let Through"





Fuse Applications

Types of circuits to be protected:

- Those with non-varying loads - normally protect against overload as well as a short circuit.
- Those with widely varying loads, e.g. DOL motors with high inrush compared to normal rating, cranes, etc. Here, fuses generally protect against short circuit only.

General Benefits and Limitations

Benefits:

- **Serves two purposes:** fault detector and interrupter
- Main virtue - speed

Limitations:

- Fixed current/time characteristic
- Needs replacing after operating
- Use for LV and MV applications (up to 66 kV)

Isolation, Switching and Tripping

Isolation:

- Basic function intended to cut off for reasons of safety the supply from all, or a discrete section, of the installation by separating the installation from every source of energy.

Switching:

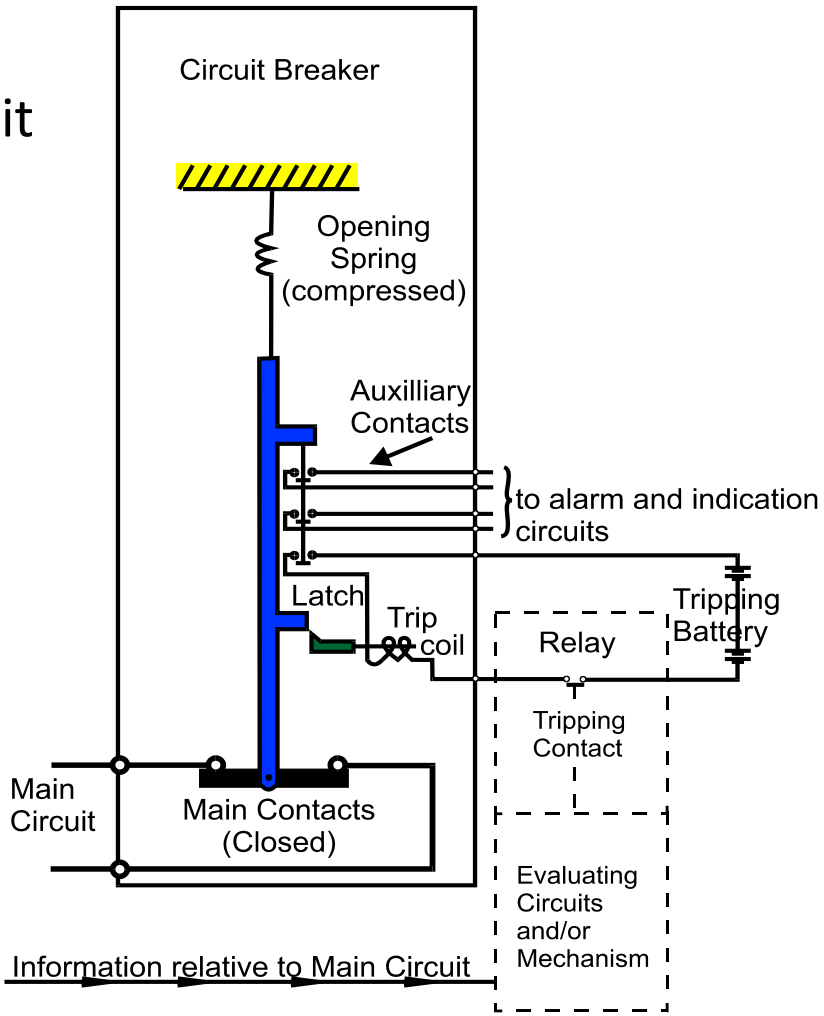
- To make or open an electric circuit or to divert load from one circuit to another carrying hundreds of amperes.

Tripping:

- Automatic trip through the operating mechanism in case of a short circuit or overload condition.

Relay-Circuit Breaker Combination

- When fuses are unsuitable or inadequate, protective relays and circuit breakers are used.
- The relay measures and determines when to open the CB.
- A wound spring normally provides the energy to operate the CB.



https://switchgearcontent.com/2019/05/14/465/relays-and-circuit-breaker-combination/

Arc Behaviour during Breaker Operation

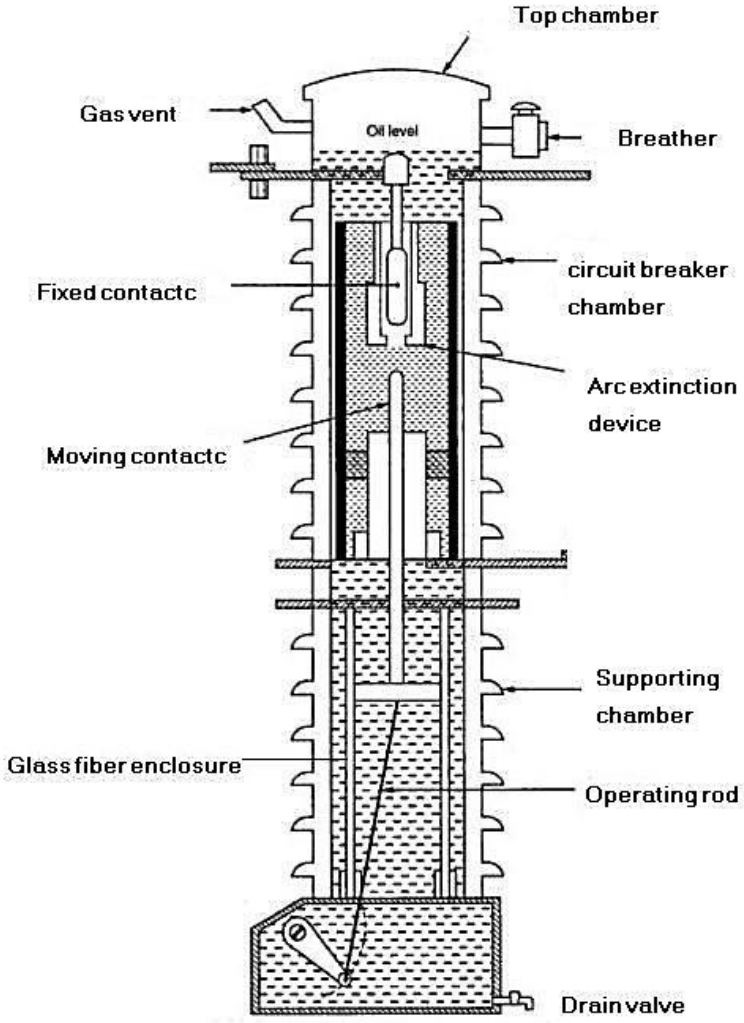
- Contacts start to separate.
- Arc forms between separating contacts resulting in ionisation of medium between contacts - *high current phase*
- Current flow interrupts at first current zero crossing. Arc quenching medium cools ionised gases - *thermal phase*
- Transient recovery voltage develops across breaker contacts. Cooling column of medium replaces hot gases and insulates contacts again preventing a re-strike - *dielectric phase*

MV/HV Circuit Breaker Types

- Circuit breakers basically classified in MV and HV power systems in terms of the dielectric used to quench the arc when the breaker opens:
  - Oil circuit breakers – bulk oil and minimum oil (MV)
  - Air blast circuit breakers (HV)
  - SF_6 circuit breakers (MV and HV)
  - Vacuum circuit breakers (MV)
- Types 1 and 2 are still used but no new installations are carried out with them
- Types 3 and 4 are now exclusively installed in new systems.

Minimum Oil Circuit Breakers (MOCB)

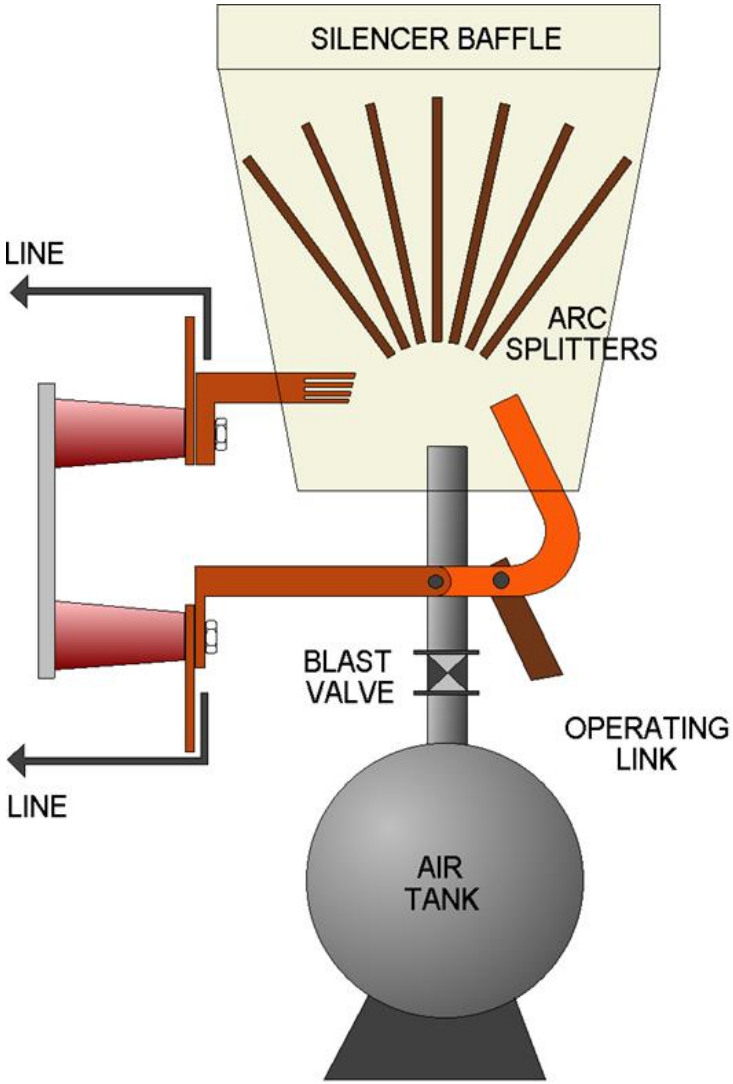
- Many breakers of this type are still in service though no longer manufactured
- Lower oil volume than bulk oil designs
- Only the oil in the upper part of the column extinguishes the arc
- Limited ratings, now obsolete design



https://www.yourelectricalguide.com/2017/09/low-oil-circuit-breaker.html

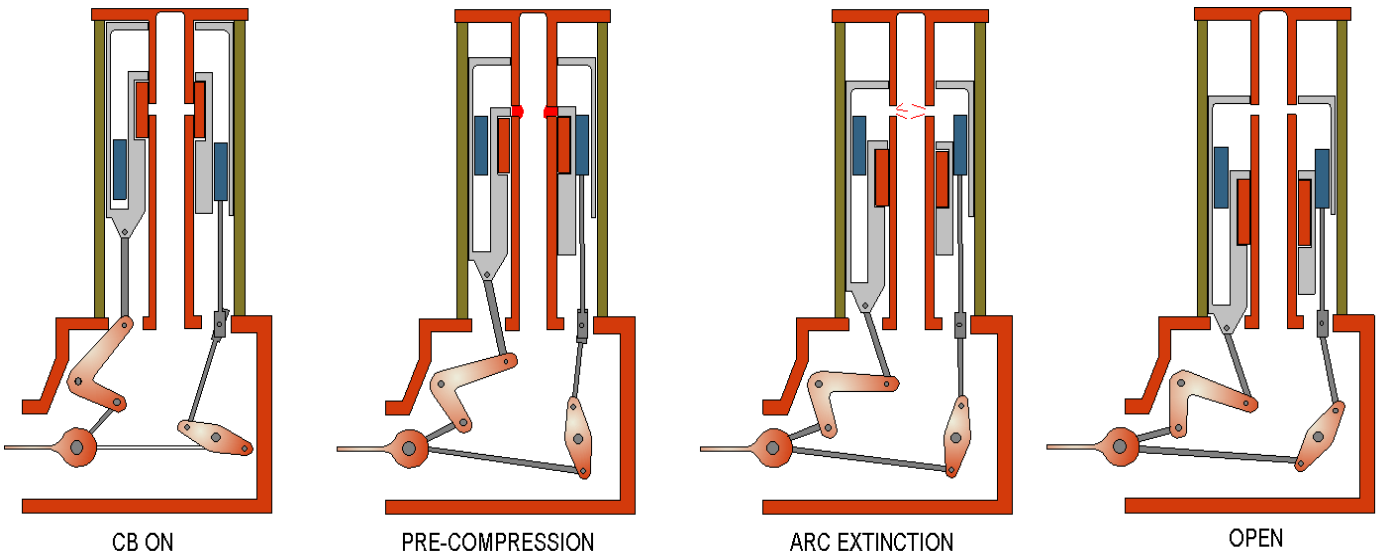
Air Blast Circuit Breakers (ABCB)

- As contact opens and arc is struck the heat causes the arc to rise towards the arc splitter.
- High pressure air then blows the arc through the splitters cooling and quenching the arc.
- High maintenance needs of the extensive compressed air plant.
- Now obsolete design.



SF_6 Circuit Breakers

- Widely used for voltages greater than 24 kV.
- SF6 breaker uses the principle of self-compression puffer type piston principle, involving relatively low over voltages across its terminals.



SF_6 Breaker Advantages and Disadvantages

Advantages:

- Soft switching
- Suitable for high voltages
- Monitoring systems in place (pressure)
- Insulation medium presents no fire hazard
- SF_6 sealed for switchgear life
- Compact system

Disadvantages:

- Fewer operations compared to vacuum but vacuum is limited to MV only
- Generally more expensive than vacuum
- SF_6 is a greenhouse gas

Vacuum Breakers Advantages and Disadvantages

Advantages:

- Large number of operations
- Compact
- Relatively inexpensive

Disadvantages:

- Cannot monitor vacuum
- Generally limited to 36 kV

Comparison of Circuit Breakers

Feature \ Insulating Medium	Air	Air blast	Oil	SF_6	Vacuum
No. Of operations	Medium	Medium	Low	Medium	High
Breaking ability	Good	Good	Good	Good	Good (Fair for soft break)
Monitoring of medium	No medium	Automatic	Manual	Automatic	Not possible
Fire hazard risk	None	None	High	None	None
Health hazard risk	None	None	Low	Low	None
Economical voltage range	Up to 1kV	33kV to 750kV	33kV to 110kV	3.3kV to 800kV	3.3kV to 36kV

End of Webinar 5

Any questions or comments?